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Master Thesis

Investigating the Application of Small Language Models for Educational Data Storytelling

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"Esto quod audes"

Abstract

Data storytelling can improve student engagement, but is often limited by the high cost of large language models. Small language models offer a cheaper option, yet their ability to create accurate and engaging educational content is unclear. This thesis studies whether SLMs can generate story-based learning materials from existing curriculum content. We designed a modular retrieval-augmented generation system with two stages: knowledge extraction and story generation. Seven Retrieval-Augmented Generation-based SLMs and 49 model combinations were tested on a custom dataset. Results show that SLMs can create high-quality educational materials when tasks are divided by model strengths. Knowledge extraction models like Phi4-14B ensured factual accuracy, while models like Qwen2.5-7B performed better in narrative generation. Pipelines using different models for each stage outperformed single-model systems. These findings confirm that, with thoughtful design, SLMs are a practical and accessible tool for educational data storytelling.

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1

Introduction

In today's era of big data, the ability to turn raw information into clear and meaningful stories has become a key skill in many elds. This practice is called data storytelling (1). It goes beyond static charts and reports by linking data points into a structured narrative. Such a narrative provides context, explains meaning, and captures the audience's attention. In education, data storytelling has great potential (2). It can simplify complex ideas, make abstract concepts easier to understand, and help students learn more e ectively. By presenting knowledge as a story, teachers can improve both knowledge retention and classroom engagement.

While LLMs dominate AI-driven content generation, their computational cost limits educational applications a gap where Small Language Models (SLMs) may o er a practical alternative. However, LLMs also have clear drawbacks: they are very large, require heavy computation, and cost a lot to run. These problems make them di cult to use in many schools and universities (3). To overcome these limitations, researchers have started to focus on Small Language Models (SLMs). These models have fewer than 15 billion parameters and are much lighter and cheaper to run. They can still generate high-quality text while using far fewer resources.

Although LLMs have been widely studied, there is little research on using SLMs for educational data storytelling. This task demands three key elements: factual precision, logical coherence, and pedagogical intuition. The goal of this thesis is to explore whether SLMs can be used to create structured, story-based learning materials from existing educational content.

This study investigates the problem in a systematic way, guided by the following research questions.

1.1 Research Questions

This study addresses the following research questions:

1. RQ1: To what extent can Small Language Models be used to generate coherent and useful educational data stories from structured materials?
2. RQ2: How do di erent Small Language Models, and their combinations in a multi-step pipeline, compare in performance, factual accuracy, and narrative quality?

RQ1 examines whether SLMs can follow instructions, stay factually correct, and produce narratives suitable for the classroom. And RQ2 aims to nd the strengths and weaknesses of di erent SLMs and combinations, and to see which setups work best for knowledge extraction and story generation.

1.2 Contributions

To answer these questions, this thesis makes the following contributions:

1. We design and implement a modular end-to-end system for educational data story-telling using Small Language Models. The pipeline uses Retrieval-Augmented Generation (RAG) and a chained-prompting strategy optimized for SLMs.
2. Our system performs a large-scale evaluation of seven SLMs within the system, and shows how the knowledge extraction and story generation stages a ect each other and provide clear guidance for choosing the best models.

The source code for this project is available at the GitHub repository[1]

1.3 Thesis Structure

The rest of this thesis is organized as follows. Chapter 2 introduces key concepts such as Retrieval-Augmented Generation, Small Language Models, and Data Storytelling. Chapter 3 reviews related research on Large Language Models in education, prompt engineering, storytelling, and RAG applications, highlighting research gaps. Chapter 4 describes the design of the RAG-based data storytelling system, including knowledge extraction and story generation modules. Chapter 5 presents evaluation results comparing di erent Small

[1] https://github.com/yip-jm/eduDS

Language Models and their combinations. Chapter 6 discusses the findings, limitations, and future research directions. Chapter 7 concludes by summarizing the contributions and key insights.

2

Background

2.1 Retrieval-Augmented Generation

Retrieval-Augmented Generation (RAG) is a technique that combines information retrieval with generative models (4), enabling large language models (LLMs) to incorporate authoritative external knowledge before generating responses. By leveraging this external information, RAG signi cantly improves the relevance, accuracy, and utility of generated content especially in specialized or knowledge-intensive tasks while e ectively mitigating hallucination issues without requiring model retraining (5, 6). This approach not only maximizes the potential of LLMs but also broadens their applicability across a wide range of professional domains.

2.1.1 Architectures and Developments

The RAG architecture primarily consists of three key components: indexing, retrieval, and generation. In the indexing stage, documents are divided into chunks, which are then transformed into vectors using an embedding model and stored in a vector database. During retrieval, the user's query is compared against the vector database to nd the top-K most relevant chunks based on similarity. In the generation stage, the retrieved content is combined with the query and input into LLMs to produce the nal answer.

According to its development, Gao et al. categorize RAG into three stages: Naive RAG, Advanced RAG, and Modular RAG(6). Naive RAG follows the traditional pipeline, consisting of indexing, retrieval, and generation. Advanced RAG enhances the retrieval process by introducing pre-retrieval and post-retrieval process(7, 8, 9), aiming to improve retrieval e ciency and increase the utilization of retrieved chunks. Modular RAG decouples components such as retrieval and generation, with each module responsible for a speci c

function. In the Modular RAG designed by Gao et al., an orchestration mechanism is introduced to dynamically select the subsequent steps in the RAG work ow(10).

2.1.2 Optimization

The optimization of RAG can be divided into three key stages based on indexing, retrieval, and generation, speci cally involving input enhancement, retriever enhancement, and generator enhancement.

Input enhancement primarily includes two methods: query transformation and data augmentation. Query transformation involves generating pseudo-documents based on the original query and using them as new inputs for retrieval, thereby improving retrieval accuracy (8). Data augmentation preprocesses the data before retrieval, where structured representations enhance the precision and usability of the information (11).

For retriever enhancement, multiple approaches can be employed to improve the quality of retrieved content and to better utilize in-context learning capabilities, including Recursive Retrieval, Chunk Optimization, Finetuning the Retriever, Hybrid Retrieval, and Reranking. Recursive retrieval performs multiple rounds of search to obtain richer and higher-quality content. Chain-of-Thought (CoT) reasoning in ReACT enables multi-step retrieval through query decomposition, allowing access to more comprehensive information (12). Chunk optimization techniques improve retrieval results by adjusting the size of text chunks. Sliding window is an e ective method that captures contexts to enhance retrieval relevance, while LlamaIndex organizes documents into a tree structure and performs hierarchical retrieval, automatically merging results layer by layer (13). As the core component of a RAG system, the retriever can further enhance its representation capabilities in specialized domains by using high-quality embedding models such as BGE and AngIE (14, 15). Hybrid retrieval integrates sparse and dense retrievers to handle diverse query types better, while reranking the retrieved results increases the visibility and utility of critical information (16).

In the generation stage, prompt engineering can guide large language models to produce more relevant and higher-quality outputs. Speci cally, this includes zero-shot, few-shot, role prompt, instruction prompt, and chain-of-thought (CoT) prompting methods. Zero-shot prompting refers to using a prompt without providing any examples, allowing the large language model to perform the task directly (17). This approach is also considered one of the key settings for evaluating the inherent capabilities of large language models. In Few-shot prompting, few-shot examples provide large language models with in-context learning, guiding them in both the format and content of the desired output (18). Role prompting

assigns the model a speci c role, which helps enhance its performance on targeted tasks by aligning its responses with the expectations of that role (19). Compared to a basic prompt, instruction prompting provides the model with a comprehensive description of the task, enabling it to generate outputs that are more precise and relevant (20). CoT prompting encourages the model to generate intermediate reasoning steps or a chain of thought, simulating the human problem-solving process to enhance its ability to perform complex reasoning tasks (21).

2.1.3 Evaluation

For the evaluation of the RAG system, it can be primarily divided into two aspects: retrieval, generation. The retrieval component is assessed based on how e ectively it identi es and ranks relevant documents from a large corpus, which is crucial for grounding the generated content in factual evidence. Metrics such as Recall, E ective Information Rate (EIR), and normalized discounted cumulative gain (nDCG) are commonly used to evaluate retrieval quality. On the other hand, the generation component is evaluated by examining the quality, relevance, coherence, and factual consistency of the nal output(22, 23).

2.2 Small Language Models

Large Language Models (LLMs) have demonstrated impressive performance across a wide range of tasks. However, their massive parameter sizes and high computational demands present signi cant challenges in terms of time and computational cost. To address these challenges, Small Language Models (SLMs), which typically have fewer than one billion parameters, have emerged as lightweight alternatives. With lower computational overhead and strong practicality, SLMs are becoming an increasingly important choice for real-world applications.

The development of SLMs relies on a variety of technical methods to reduce model size while maintaining strong performance. Knowledge distillation allows smaller models to learn from the outputs or internal representations of larger models, enabling e ective domain-speci c enhancement (24). Weight pruning reduces computational overhead by removing redundant parameters without signi cantly sacri cing accuracy (25). Quantization uses low-precision numerical formats to lower memory consumption (26). E cient network architectures such as DistilBERT (27) and MobileBERT (28) achieve reduced model size and improved computational e ciency through techniques like parameter sharing and bottleneck layers, while preserving model depth.

As shown in Table 2.1, in the open-source community, several well-optimized models have emerged, including DeepSeek-LLM, Gemma, Qwen2.5, OpenChat, Llama3.1, OLMo2, and Phi-4. DeepSeek-LLM is known for its strong capabilities in code generation and mathematical reasoning, making it well-suited for complex tasks (29). Gemma performs exceptionally well in academic benchmark tests for language understanding, reasoning, and safety (30). Llama3.1 enhances long-context retention and multilingual generalization while aligning better with safety benchmarks (31). Olmo2 is a fully open and transparent research model designed for reproducible scienti c evaluation (32). Openchat is a conversational AI model designed to enhance open-source language models using mixed-quality data, with optimizations for instruction following and multi-turn dialogue (33). Qwen2.5, trained on high-quality bilingual data with an advanced tokenizer, excels in multilingual and instruction-following tasks (34). and Phi-4, developed by Microsoft, achieves a balance between performance and e ciency through architectural optimizations and high-quality training data, demonstrating strong capabilities in reasoning and mathematics (35).

SLMs	Deepseek-llm	Gemma	Llama3.1	Olmo2	Openchat	Qwen2.5	Phi-4
Size	7B	7B	8B	7B	7B	7B	14B
Provider	Deepseek	Google	Meta	Allenai	Tsinghua	Alibaba	Microsoft

Table 2.1: Basic information of several open-source SLMs

2.3 Data Storytelling

In the era of abundant data, the ability to extract meaningful narratives from complex datasets has become a crucial interdisciplinary skill. Traditional methods of data presentation, which focus on static visualizations and numerical reports, have now evolved into a more dynamic and context-sensitive practice known as data storytelling. This approach integrates data analysis, narrative construction, and visual presentation, which not only aids understanding but also enhances engagement, persuasion, and decision-making.

Based on my previous research (36), data storytelling is typically conceptualized as a three-stage process comprising data exploration, story construction, and story narration. The exploration phase involves descriptive, diagnostic, predictive, and prescriptive analytics, enhanced by exploratory data analysis. Story construction entails two key components: narrative modeling and visual representation. A variety of narrative frameworks, such as Freytag's Pyramid, Aristotle's three-act structure, and Campbell's Hero's Journey, are used to shape coherent, emotionally resonant stories. Visual representation must balance

clarity, salience, and audience engagement, often guided by design heuristics rooted in human visual perception. The nal stage, narration, focuses on the delivery of the story, ranging from linear, author-driven formats to interactive, reader-driven designs. AI is increasingly integrated into this stage, enabling adaptive storytelling, personalized content, and immersive environments through AR/VR technologies.

Applications of data storytelling are notably domain-speci c, with signi cant methodological and strategic variations observed across journalism, education, nance, and medicine, as shown in Table 2.2. In journalism, data storytelling is used to communicate complex societal issues to the general public, employing structures like the 5W-1H model and integrating interactive visualizations to enhance comprehension and engagement. In education, digital storytelling serves as a pedagogical tool, emphasizing emotional connection and student co-creation through multimedia narratives. In nance, storytelling frameworks are designed to support decision-making, with curated dashboards and structured reports aimed at di erent stakeholders. In medicine, narrative and visual techniques are applied to communicate clinical data and treatment information e ectively to both professionals and patients, often employing 2D/3D interactive tools to enhance accessibility and comprehension.

Applications	Journalism	Education	Finance	Medicine
Data Exploration	descriptive	abstract concepts	descriptive, diagnostic, predictive	descriptive, diagnostic
Story Construction	linear structure; 5W-1H framework	7 key elements, 4 key steps, student participation	5 key attributes, three-stage policy model	characters, plot, themes, Freitag's Pyramid, Hero's Journey
Visual Representation	annotated charts, interactive graphics, multimedia elements	multimedia integration (images, audio, video)	line charts, dynamic graphs, dashboards	interactive 2D/3D visualizations, medical animations
Story Narration	linear/hybrid structures	rst-person perspective, question-driven	problem-solution structure, tailored to audience	patient-centered, reader-driven model, interactive slides

Table 2.2: Comparison of Data Storytelling Methods Across Di erent Applications

Recent advancements in arti cial intelligence have profoundly reshaped the landscape of data storytelling. AI technologies now contribute not only to automated data analysis but also to the generation of narrative content, visualization recommendations, and delivery customization. Despite these innovations, scholars emphasize that human oversight remains indispensable to ensure contextual appropriateness, narrative coherence, and ethical integrity, particularly as AI systems may lack domain-speci c understanding and transparency. Therefore, the eld is moving toward a paradigm of human AI collaboration,

where automation enhances, rather than replaces, human interpretive and communicative capacities.

The literature study consistently conceptualizes data storytelling as a three-stage process: data exploration, story construction, and story narration, supported by both narrative theory and visual design principles. While domain-speci c adaptations exist, such as the participatory and multimedia-rich approach in education, current research still shows several gaps relevant to this thesis: (1) Most educational applications prioritize narrative engagement but lack systematic frameworks to ensure factual accuracy and structured knowledge delivery, particularly when transforming abstract curriculum concepts into coherent stories. (2) The integration of AI, especially Small Language Models, remains underutilized for end-to-end educational storytelling pipelines. (3) Although human AI collaboration is widely acknowledged as necessary for contextual appropriateness and ethical integrity, practical, scalable systems embedding retrieval-augmented generation into storytelling for classroom use are scarce. This thesis aims to address these gaps by building a modular RAG-based system that brings together factual accuracy, narrative coherence, and alignment with teaching goals, using SLMs to create accessible and high-quality educational data stories.

3

Related Work

Large Language Models (LLMs) have demonstrated signi cant potential in various areas, particularly in education. They can create content, provide personalized feedback, and assist in building interactive learning environments. Due to these abilities, numerous studies have been conducted in this eld. In this chapter, we review previous research from four main areas that are closely related to our topic: (1) how LLMs are used in education, (2) prompt engineering methods for educational purposes, (3) using storytelling and narratives in teaching, and (4) Retrieval-Augmented Generation (RAG) in smart educational systems. At the end, we point out the research gaps that our study hopes to ll.

3.1 Applications of LLMs in Education

The integration of LLMs into educational settings has enabled novel teaching and learning paradigms. George (37) outlined four key areas where generative AI can bene t postgraduate education: personalized learning, automated feedback, intelligent research support, and content creation. However, the study remains largely exploratory and lacks a concrete system architecture for real-world deployment. In contrast, our work o ers a fully modular and deployable framework tailored for classroom integration, focusing on practical usability rather than conceptual potential.

Similarly, Thüs et al. (38) developed a RAG-based system OwlMentor, which demonstrates how AI can support metacognitive strategies (such as self-questioning, summarizing, and self-explanation) and improve answer accuracy through semantic retrieval and context generation. In contrast, our work investigates the use of small models for adaptive, instructor-driven teaching. We enable teachers to integrate their own course materials

and leverage data storytelling to interpret students' learning trajectories and conceptual understanding.

3.2 Prompt Engineering Strategies for Educational Tasks

Prompt engineering plays a critical role in adapting LLMs to educational goals. Chen et al. (39) conducted a systematic review of prompt strategies used in K 12 STEM education, categorizing techniques such as zero-shot learning, chain-of-thought prompting, and retrieval-augmented prompting. These methods were evaluated across multiple tasks, including tutoring, curriculum alignment, and concept explanation, with generally positive outcomes.

Building on these ndings, our system integrates prompt engineering as a core design element. We develop tailored prompt templates aligned with instructional goals and apply them systematically across stages of content generation, from factual retrieval to narrative synthesis. This integration improves the factuality, uency, and educational relevance of outputs, even when using smaller, more e cient language models.

3.3 Storytelling and Narrative Generation in Education

Storytelling has long been recognized as an e ective pedagogical method to improve student engagement and conceptual understanding. Jiang et al. (40) applied narrative generation techniques in legal education, using LLMs to simplify complex legal concepts for non-experts. Their results showed improved comprehension through narrative framing, although their approach remained con ned to a speci c domain.

Our work generalizes this idea by introducing a storytelling pipeline adaptable to STEM education. Through the use of guided prompts and model sca olding, we transform structured knowledge into pedagogically e ective teaching stories. Furthermore, we evaluate several small-scale LLMs in terms of narrative uency and factual accuracy to ensure that the generated content aligns with instructional goals.

In addition, Eldan and Li (41) demonstrated that small LLMs, when guided with carefully designed prompts, are capable of producing coherent and uent short stories. Inspired by their ndings, our system incorporates prompt templates, prede ned structural cues, and content constraints to support smaller models in generating instructional narratives that are both imaginative and educationally meaningful.

3.4 RAG in Educational Systems

RAG has emerged as a key mechanism for combining external knowledge retrieval with generative outputs, enhancing both factual accuracy and contextual relevance. Its use in education is growing, particularly for question answering and reading support. Wang et al. (3) provided a comprehensive review of LLMs in education, examining use cases such as intelligent tutoring, automated grading, question generation, and content summarization. Of particular relevance to our work is their identification of RAG as a promising approach to enhance the factual reliability of LLM outputs in educational contexts. While their study highlights the potential of such methods, our work advances this discourse by operationalizing RAG within a structured lesson generation pipeline, with a specific focus on the creation of story-based instructional content that aligns with pedagogical goals.

For instance, the AI-University platform (42) leverages LLMs and RAG to align generative content with instructional goals in scientific education, improving alignment with learning standards and domain knowledge. While the system showcases the utility of RAG for educational support, it primarily enhances isolated tasks rather than generating structured content such as lesson plans or stories.

Our work expands the role of RAG from a supportive backend function to a central design component. We use RAG not only to ensure factual accuracy but also to structure content progression and guide the generation of coherent, multi-part teaching narratives.

Furthermore, Pan et al. (43) introduced a method to guide generative storytelling using structured knowledge graphs. Their system improved coherence and topic relevance in generated narratives, while it may be less suitable for introspective, emotion-driven stories, particularly when using smaller LLMs.

3.5 Summary and Research Gaps

Across these thematic areas, prior research highlights the significant potential of LLMs in education. However, several gaps remain:

- Lack of practical, scalable systems : Many studies stop at proof-of-concept systems or domain-specific tools, without addressing real-world classroom usability or modularity.
- Underutilization of storytelling : Despite its pedagogical value, storytelling is rarely treated as a central design component in educational AI systems.

- Prompt engineering fragmentation : Prompt design is often addressed in isolation, without integration into full pipelines for educational content generation.
- Limited role of RAG : Although increasingly used for factual enhancement, RAG is seldom embedded as a core mechanism driving structured educational outputs.
- Over-reliance on large models : Many systems rely on high-resource LLMs, limiting accessibility and scalability.

Our work addresses these challenges through an end-to-end, modular system that integrates RAG, prompt engineering, and small language models to generate educational stories grounded in curriculum-aligned knowledge. By embedding storytelling and factual retrieval within a guided prompting framework, we o er a novel, lightweight solution for real-world educational deployment.

4

Design

This chapter presents the design of the proposed RAG-based data storytelling system for educational applications. It introduces the overall architecture and describes each major component in detail, including the knowledge extraction module, the story generation and narration module, the dataset construction process, and the evaluation methodology. Additional details, including step-by-step implementation guidelines and recommendations for adapting the system to di erent contexts, are provided in Section 6.3.

4.1 Top-level Design

The proposed system is designed as a complete pipeline for generating educational data stories based on structured domain knowledge. It combines RAG with modular story generation and interactive narration, aiming to support teachers in preparing lesson content that is both accurate and engaging. The system's architecture, shown in Figure 4.1, is modular and interpretable, consisting of three main components:Knowledge Extraction , Story Generation , and Story Narration .

Figure 4.1: The Top-level Design of RAG-based Data Storytelling System

4.2 Knowledge Extraction

The knowledge extraction module is the first and foundational step in the overall system architecture. Its objective is to convert raw learning materials, such as textbooks and PDF documents, into semantically meaningful, structured chunks of information suitable for downstream story generation and interactive narration. This section details the design choices and work flow of the knowledge extraction process, as illustrated in Figure 4.2

Figure 4.2: Knowledge Extraction in Detail

4.2.1 Document Processing and Chunking

The process begins by collecting input documents into the system's knowledge base. These documents are loaded using a directory loader and processed through a two-stage chunking pipeline, which is designed to maintain both semantic clarity and proper text length.

To deal with the complexity of educational documents such as multi-column layouts, tables, and figures we use an open-source tool called minerU for PDF parsing. minerU performs well in preserving the original document structure and ensures clear content extraction.

After parsing, the documents go through a dual-stage segmentation process:

- Semantic Chunker: First, the documents are split into semantically meaningful parts using embedding-based similarity. This ensures that each chunk focuses on one main idea or concept.
- Recursive Character Text Splitter: Then, a second layer of segmentation is applied. This uses a character-based recursive approach to keep chunks within a length limit, making them suitable for downstream processing.

Each chunk is then tagged with metadata, including a unique chunker_id and an inferred content_type, which helps in tracking and retrieving relevant information later.

4.2.2 Hybrid Information Retrieval

To support RAG, the system implements a hybrid retrieval strategy that integrates both keyword-based and semantic search mechanisms. This design is motivated by the strengths and limitations of each method:

- Keyword-based retrieval (BM25): This method is effective for finding exact matches, especially when the query contains technical terms from the educational domain.
- Vector-based retrieval (Chroma + HuggingFace Embedding): This allows the system to retrieve semantically similar content, even when the query uses different words or phrasing.

The results from these two retrieval methods are merged by a Hybrid Retriever, which ranks them together to improve accuracy and coverage.

4.2.3 Re-ranking via Cross Encoder

After retrieval, all candidate chunks are passed into a Cross Encoder for re-ranking. Unlike the individual scoring methods used in sparse or dense retrieval, the Cross Encoder processes the query and chunks together, which improves the accuracy of relevance scoring.

Only the top-k documents with the highest relevance scores are retained for the generation phase. This step ensures that only contextually appropriate and high-quality content is used during narrative construction.

4.3 Story Generation and Narration

The story generation and narration component is responsible for transforming structured domain knowledge into engaging, pedagogically e ective educational stories. Given the token limitations of SLMs, our system does not generate the entire story in one prompt. Instead, it adopts a modular pipeline approach using chained prompts as shown in Figure 4.3. This strategy allows complex tasks to be broken down into smaller, more manageable steps, ensuring higher quality and coherence in the generated output. The overall design follows four core principles:

- Decoupling: The system rst builds a general lesson outline, and teachers can then request detailed story modules as needed.
- Mapping & Transformation: Structured knowledge elds are carefully mapped to story elements based on storytelling theory.
- Interactive Generation: Teachers are actively involved. They can review, select, and adjust each generated output step by step.
- Focus & Brevity: Each prompt is task-speci c, keeping the output clean, relevant, and within the token limits of small models.

4.3.1 Story Generation

The system employs a chained, modular prompt strategy: each teaching module is created through two separate prompts one for storytelling and another for activity generation both centered around a single concept. This strategy follows the design principle of divide and conquer , reducing cognitive load for the model and ensuring focus and depth at every step.

For each Core Concept in the knowledge base, the system rst generates a micro-narrative using a carefully designed storytelling prompt. The prompt explicitly follows a three-part narrative structure: Problem Solution Impact, which aligns with Abrahamson's(1) narrative theory. In addition, elements from Alismail's(2) digital storytelling framework such as Dramatic Question and Point of View are embedded as storytelling hooks to increase emotional resonance and student engagement.

The prompt only includes data from one Core Concept at a time (e.g., its de nition, signi cance, strengths, and weaknesses). This focused context allows the model to fully

Figure 4.3: Flowchart of Story Generation and Narration Design

explore the concept, ensuring that the generated story is both detailed and educational. Each story is formatted in Markdown and labeled with a clear section title.

4.3.2 Narration and Interaction Design

Immediately after generating the story, the system proceeds to create interactive classroom activities for the same Core Concept. These activities are designed to reinforce key ideas and stimulate discussion, making the story more than just passive content.

The activity generation prompt asks the model to create:

- Debate questions based on strengths and weaknesses of the concept;
- Real-world scenarios where students must make decisions or solve problems.

This step operationalizes the concept's educational potential through interactive narration, following the principles of learner-centered pedagogy. Activities are designed to be adaptable for group discussions, in-class polls, or homework assignments.

Each activity section is generated separately and appended under its corresponding story module, forming a complete mini-package.

4.3.3 Lesson Guide and Classroom Use

Once both the story and its corresponding activity are generated for a concept, they are combined into one Markdown section. Over multiple iterations, this process builds a complete, modular lesson package, with each concept turned into an independent teaching module. These modules can be rearranged, removed, or reused depending on the teacher's needs.

This design brings three major bene ts:

1. Instructional clarity: Each concept's story and activity are paired, making the material easy to teach and follow.
2. Token e ciency: Since each module is generated separately, there is no danger of running out of model context.
3. Personalization potential: Teachers can manually adjust individual modules or regenerate speci c parts to suit di erent audiences.

Overall, the story generation and narration system bridges the gap between structured knowledge and educational storytelling. By adopting a modular, controllable design, teachers can exibly guide the generation process while producing high-quality narrative outputs that are engaging, informative, and classroom-ready. The modular structure and interaction design further support meaningful learning experiences that extend beyond simple content delivery.

4.4 Dataset Construction

To support the development of a RAG-based data storytelling system in the educational domain, a high-quality dataset is essential. The goal of this dataset is to simulate realistic educational question-answer scenarios that align with actual teaching processes. We designed and generated a custom dataset tailored to classroom teaching and curriculum planning. The motivation for creating this dataset is threefold:

1. Educational Relevance: Existing datasets often lack structured teaching context or fail to align with knowledge points taught in actual courses. Our dataset directly re ects curriculum planning and instruction design around speci c topics.

2. Structured Knowledge Extraction: Our system relies on the accurate identi cation of core concepts and their contextual importance. Therefore, our dataset is annotated with rich concept-level metadata for downstream reasoning and storytelling.

3. Realistic Simulated Data: Given the limited availability of labeled educational data, we simulate a teacher's thought process using GPT-4o to generate question-answer pairs that re ect real-world use cases automatically.

The dataset consists of 100 structured question-answer (QA) pairs, each centered around a speci c knowledge topic, designed to simulate a teacher's lesson preparation process.

To generate realistic and domain-relevant content, we designed a semi-automated pipeline using GPT-4o. In this pipeline, the model was guided to act as a teacher preparing for a class session on a selected topic. The model received detailed instructions to simulate a planning query typically representing a teacher requesting a breakdown of speci c subtopics, conceptual comparisons, or instructional focus areas. These prompts closely mirror real-world teaching needs in computer science education.

For example, one such prompt was:

{ "Question": "I need to prepare lessons on virtualization, with a focus on full virtualization, para-virtualization, and hardware-supported virtualization. Include how each method works, the role of hypervisors (Type 1 and Type 2), and performance implications." }

From this input, the model generates a structured response that adheres to a strict schema, including the high-levelknowledge topic , core concepts with detailed metadata, and a concise summary , as shown in Table 4.1. This schema design ensures consistency and completeness across all generated entries.

Field Name	Description
Question	The original user query simulating a teacher's request for preparing a lesson on a speci c topic.
Knowledge_Topic	A high-level subject area that the question falls under, such as Virtualization or Mutual Exclusion Algorithms
Core_Concepts	A list of key concepts relevant to the question, each annotated with detailed semantic metadata (see below).
Concept	The name of the concept involved in the topic.
Definition	A concise explanation of the concept.
Key_Points	A list of major facts or ideas that de ne the concept, ideally three or more.
Significance_Detail	Explanation of why the concept is important or relevant in context.
Strengths	One or more strengths or bene ts of the concept.
Weaknesses	One or more limitations or challenges associated with the concept.
Source_Context	Relevant background content supporting the question.
Overall_Summary	A one-to-two sentence summary that provides a direct, concise answer to the question.

Table 4.1: Schema of a QA Pair in the Dataset

While the topics are grounded in foundational and commonly taught areas of computer science, our focus is not on exhaustive topic coverage but rather on demonstrating the feasibility and e ectiveness of structured QA generation for downstream storytelling tasks.

Importantly, the dataset was constructed with a forward-looking objective: to serve as

a foundational layer for generating rich, accurate, and pedagogically grounded educational stories. Each QA pair provides both the content and structure necessary for later stages of narrative construction with small language models, such as identifying causal relationships, sequencing ideas, and selecting relevant conceptual elements for data storytelling.

These datasets yield a total of 700 results in the RAG module (100 per language model) and a total of 4900 results in the Story Generation module.

4.5 Evaluation

To evaluate the system's generalizability and e ciency across di erent model scales and capabilities, we conducted a series of comparative experiments involving seven leading small language models: DeepSeek-LLM, Gemma, Qwen 2.5, OpenChat, LLaMA 3.1, OLMo 2, and Phi-4. These models were tested on identical input knowledge bases, using the same prompt templates and orchestration logic. We conducted a two-part evaluation focused on its two main functional stages: (1) Knowledge Extraction and (2) Story Generation and Narration.

4.5.1 Knowledge Extraction Evaluation

The goal of the Knowledge Extraction module is to convert educational PDF materials into a structured and machine-readable knowledge base in JSON format. The quality of this stage directly a ects the reliability of all downstream story generation tasks. All evaluation metrics are scored on a scale from 0 (worst) to 5 (best).

We designed two automatic metrics to evaluate the knowledge extraction results:

1. Retrieval Score: This score measures how well the retrieved chunks semantically align with both the user's question and the key information in a reference set. For each reference sentence and the input question, we compute its maximum similarity with the retrieved chunks, and take the average across all.

2. Generation Score: This score assesses the quality of the generated answer, including faithfulness, accuracy, and completeness. We de ne three sub-criteria:

   - Faithfulness: We check whether the generated content is strictly grounded in the retrieved context.
   - Accuracy: This measures whether all key facts, concepts, and de nitions are correct.

- Completeness: This evaluates whether all required aspects of the question are addressed. For example, in our dataset, the output should cover de nition, signi cance, strengths, and weaknesses.

$$\text{Generation Score} = \frac{\text{Faithfulness} + \text{Accuracy} + \text{Completeness}}{3}$$

Together, these two scores provide a comprehensive view of the RAG system's performance. The Retrieval Score ensures that the system is using the right evidence, while the Generation Score evaluates whether the nal output is both factually reliable and well-written. To re ect their relative importance, we compute an overall score as a weighted sum:

$$\text{Overall Score} = 0{:}4 \quad \text{Retrieval Score} + 0{:}6 \quad \text{Generation Score}$$

This weighting emphasizes the quality of the generated answer while still accounting for the relevance of the retrieved evidence. In many real-world scenarios, users primarily care about the nal answer's uency and factual accuracy, which often depends on but is not solely determined by the quality of retrieval.

4.5.2 Story Generation Evaluation

This stage is the most visible output of the system, where structured concepts are turned into engaging teaching stories and class activities. The evaluation here aims to assess the educational quality, logical consistency, and generation controllability across models.

We used three automatic evaluation metrics to score each generated module:

1. Story Quality Score: Each teaching story is evaluated by DeepSeek-V3 acting as a scorer. We de ned a scoring rubric with four criteria:

   - Structure: Whether the story follows the required Problem Solution Impact structure
   - Factual Accuracy: Whether the story content stays consistent with the source concept
   - Narrative Coherence: Whether the story ows logically without contradictions

- Educational Engagement: Whether it includes teaching hooks like questions or analogies
- Fluency & Consistency: Whether the text contains grammar issues or hallucinations

$$\text{Story Quality Score} = \frac{\text{Sum of 5 Metrics}}{5}$$

2. Activity Quality Score: For each classroom activity (e.g., debate, roleplay, or scenario-based discussion) generated from the concept's strengths and weaknesses, we evaluate:

   - Narrative Grounding: Is the activity meaningfully embedded in the generated story context (e.g., characters, plot tension, or moral dilemma), rather than being a generic or disconnected task?
   - Classroom Applicability Usability: Can the activity be directly used in a classroom setting to promote student engagement, critical thinking, or collaborative exploration of the concept?

$$\text{Activity Quality Score} = \frac{\text{Sum of 2 Metrics}}{2}$$

Then we have:

$$\text{Overall Score} = \frac{\text{Activity Quality Score} + \text{Story Quality Score}}{5}$$

All evaluations are performed by prompting DeepSeek-V3 with carefully designed scoring rubrics. For each task (e.g., story quality, activity engagement), we de ne 3 4 detailed sub-criteria. Each sub-score ranges from 0 (worst) to 5 (best). All nal scores are calculated as the average of their respective criteria scores.

5

Results

This chapter presents a comprehensive evaluation of seven small language models in data storytelling for educational applications. The evaluation consists of two main components: the performance of the Knowledge Extraction, which is responsible for information retrieval and factual integration, and the performance of the Story Generation, which transforms retrieved information into coherent narrative texts. The evaluation follows a set of prede ned criteria based on the proposed design in Section 4.5.

5.1 Knowledge Extraction Results

To better understand model behaviors in Knowledge Extraction tasks, we conducted a comparative evaluation across multiple models. The experimental results revealed notable di erences in performance among the models. Their detailed evaluation scores are presented in the Figure 5.1, ranked in descending order by Overall RAG Score.

Key ndings are summarized as follows:

1. Overall Performance tiers among Models

   The model Phi4_14b (3.67) achieved the highest overall score and outperformed others across all sub-metrics, establishing itself as the top performer. It was followed closely by Qwen2.5_7b (3.60). Together, these two models formed the top-performing group. In contrast, the remaining models belonged to a competitive middle tier, with minimal di erences in their overall scores (only 0.07), indicating similar levels of competence in RAG tasks. Olmo2_7b (2.78) performed signi cantly worse than the others, with an overall score nearly 19% lower than that of the next-lowest model.